

# Mandatory Physical Calibration Guide

DO NOT PRINT THE FULL ENCLOSURE YET. PRINT AND COMPLETE THE CALIBRATION PARTS FIRST.

Use the same material, nozzle, layer height, wall count, extrusion settings, bed preparation, and chamber temperature that you will use for the enclosure.

## Print first

Print these exact files from CALIBRATION\_PARTS/:

1. coupon\_official\_interface.3mf
2. coupon\_heat\_set\_insert.3mf
3. coupon\_active\_driver.3mf
4. coupon\_passive\_radiator.3mf
5. coupon\_gasket\_base.3mf
6. coupon\_gasket\_cap.3mf
7. coupon\_cable\_passage.3mf
8. cable\_gland.3mf in TPU 95A

ASA baseline: 0.4 mm nozzle, 0.20 mm layers, five walls, six top/bottom layers, 35% gyroid, 100-110 C bed, 250-260 C nozzle, enclosure closed, no supports, 5 mm brim. PETG alternative: 235-250 C nozzle, 75-85 C bed, three-hour dry cycle if stringing is present.

## Measure and enter values

Use the engraved labels and IMAGES/calibration\_\*.png.

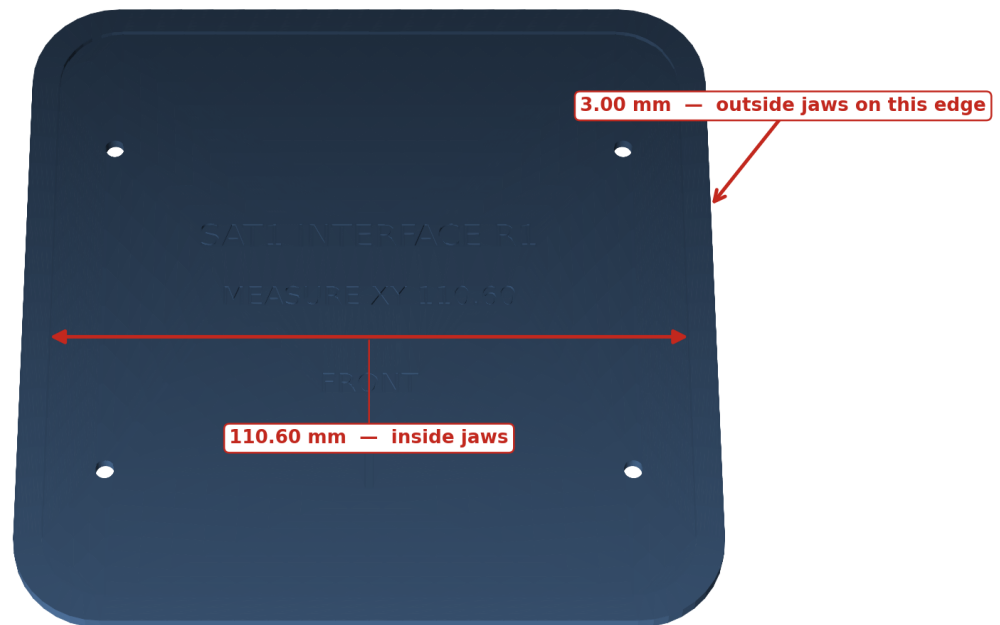
| Check        | Where/how                                                                 | Nominal           | Pass                                                            | Input key                                                     |
|--------------|---------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------|---------------------------------------------------------------|
| XY scale     | Inside jaws across<br>`MEASURE XY 110.60`<br>recess, three heights        | 110.60 mm         | 110.40-110.80 mm after<br>correction                            | `xy_scale_correction_frac<br>tion = 110.60 / measured<br>- 1` |
| Z scale      | Outside jaws across<br>clean 3.00 mm coupon<br>edge, four corners         | 3.00 mm           | 2.90-3.10 mm                                                    | `z_scale_correction_fracti<br>on = 3.00 / measured - 1`       |
| M3 clearance | Try a clean ISO M3 screw<br>in labeled 3.4/3.5/3.6<br>holes               | 3.4 mm            | smallest hole that falls<br>through without force               | chosen diameter minus<br>3.4                                  |
| Insert bore  | Install identical inserts in<br>4.0/4.1/4.2/4.3 blind bores               | 4.2 mm            | square, flush, no<br>crack/spin at 0.35 N m                     | chosen diameter minus<br>4.2                                  |
| Driver fit   | Seat the purchased<br>ND91-4 in the labeled<br>coupon                     | catalog interface | drops in by hand, <=0.30<br>mm radial play, flange lies<br>flat | cutout correction and<br>measured flange<br>thickness         |
| Radiator fit | Seat one passive radiator<br>in the labeled coupon                        | catalog interface | drops in by hand, <=0.30<br>mm radial play, flange lies<br>flat | cutout correction and<br>measured flange<br>thickness         |
| Gasket       | Tighten cap on a strip of<br>the actual sheet until both<br>stops contact | 2.00 to 1.50 mm   | 15%-45% compression;<br>no open light path                      | sheet thickness and<br>compressed-thickness<br>offset         |
| Cable gland  | Fit actual two 22 AWG<br>conductors and gland in<br>cable coupon          | 8.0 mm passage    | moderate finger force;<br>gland cannot rotate or lift           | cable-passage offset                                          |

Do not use caliper tips to measure a 3-4 mm hole; the screw and insert are the functional gauges. Enter only values you measured.

## CAD measurement illustrations

### Measure the marked slot, and the flat edge

Inside jaws across the recess. Outside jaws on any clean 3.00 mm edge.

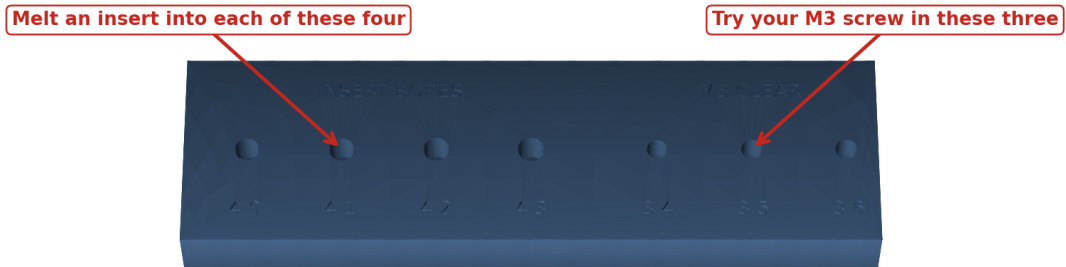


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

Official-interface XY and Z measurement

# Use the screw and the insert as gauges

Do not measure these holes with caliper tips. The hardware is the gauge.

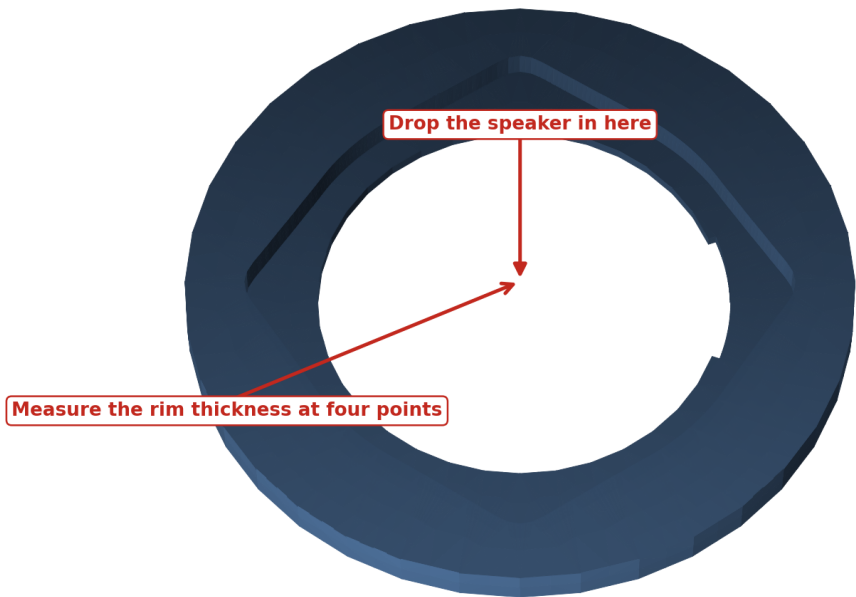


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

M3 screw and insert functional gauges

# Check your speaker drops in

It must drop in by hand and sit flat. Never force it.

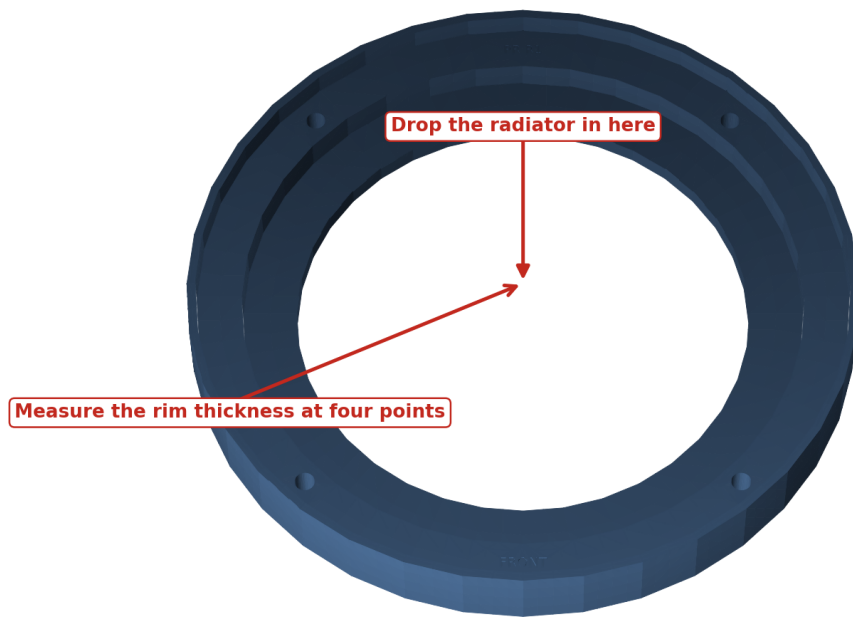


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

Active-driver coupon fit check

## Check your radiator drops in

It must drop in by hand and sit flat. Never force it.



FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

Passive-radiator coupon fit check

## Squash a strip of your foam between these two

Tighten until the two hard stops touch, then measure the gap that is left.

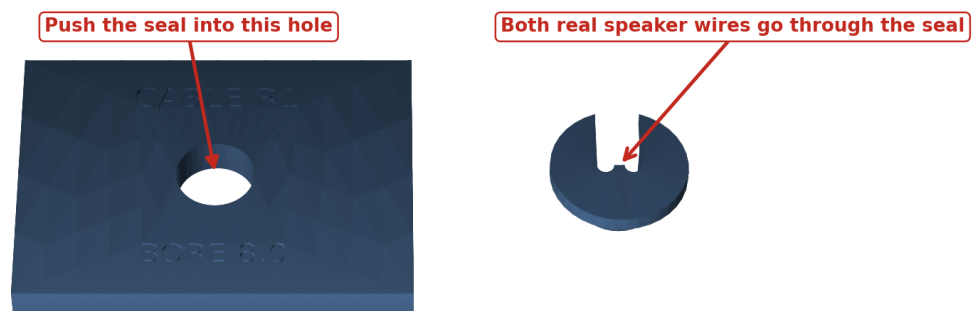


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

Gasket compression coupon stack

## Push the seal and both wires through the hole

It should take firm finger pressure, then stay put and refuse to twist.



FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

Cable passage and TPU gland check

## Edit one file

Copy `CALIBRATION_INPUT_TEMPLATE.yaml` to `config/physical_calibration.yaml`, or run:

```
python scripts/calibrate.py
```

The file exposes only user-facing corrections. Safe limits reject implausible values before any full part is built.

## Regenerate

```
make calibrated-release
```

Success ends with all validation, documentation, mutation, and package checks passing. Output is in `release/Satellite1-Ultra-RC1/`.

Example successful finish:

```
documentation PASS; 9 guides, 7 PDFs  
release/Satellite1-Ultra-RC1 (all required files present)
```

Reprint every coupon affected by a nonzero correction. You are cleared for the full enclosure only when all eight checks above pass on the corrected coupons.

## Common failures

- Warped official coupon: improve enclosure temperature, clean the bed, add the brim, and do not compensate a warped part.
- Every hole undersized: check flow and elephant-foot compensation before adding a large CAD offset.
- Insert cracks the boss: choose a larger coupon bore or reduce iron dwell; do not increase torque.
- Component will not sit flat: remove only print strings. Do not sand a sealing land; correct the cutout and reprint the coupon.
- Gland leaks or spins: verify conductor OD  $\leq 1.8$  mm, dry TPU, then correct and reprint both the coupon and gland.

Source commit `de0e76d018d5`. No physical validation has been performed.